

Comparison of base pair hydrogen bonding and base stacking interactions from QM and the MM force field parameters used in this study.

Table 1. Base-pair hydrogen-bonding minimum interaction energies and distances from QM Calculations and the MM force field parameters used in this study. QM data at the CCSD(T) complete basis set limit for hydrogen bond interactions as previously reported.¹

Complex	Distance	QM		MM	
		Interaction Energy, kcal/mol	Distance, Å	Interaction Energy, kcal/mol	Distance, Å
G:C WC	H1-N3	-28.80	1.91	-27.02	2.00
mG:mC WC	H1-N3	-28.50	1.87	-27.01	2.00
A:T WC	H3-N1	-15.43	1.82	-12.26	2.14
mA:mT H	H3-N7	-16.27	1.75	-13.11	1.89
A:C pl	H61-N3	-15.90	1.89	-16.59	1.83
G:G pl	H1-N7	-18.40	1.85	-17.42	1.80
G:A 1	H21-N1 (H1-N1)	-17.50	1.82	-18.42	2.08
G:A 1 pl	H1-N1	-16.10	1.94	-18.41	2.08
G:A 2	H21-N7	-10.90	1.96	-7.45	1.86
G:A 2 pl	H21-N7	-10.50	1.93	-7.45	1.86
G:A 3	N7-H1	-16.80	1.82	-16.13	1.87
G:A 4	N1-H21	-12.10	1.92	-8.97	1.94
A:A 1	N1-H61	-13.10	1.92	-13.18	1.88
A:A 2 pl	N1-H62	-12.30	1.95	-10.61	1.91
A:A 3 pl	H62-N7	-10.90	1.98	-9.95	1.78
	Average	-16.23	1.89	-14.93	1.93

Table 2. Base-stacking minimum interaction energies from PDB-derived ADNA and BDNA stacked geometries at the MP2/cc-pVDZ//DF-SCS-MP2-aug-cc-pVTZ model chemistry (present study) and idealized-planar stacked structures at the MP2/cc-pVDZ//DF-MP2-aug-cc-pVTZ as previously reported.²

	QM, kcal/mol	MM, kcal/mol
ADNA		
aa 1ikk 67	-5.45	-5.76
ac 1zf8 34	-8.20	-6.96
ac 2pkv 1415	-7.51	-7.64
ag 1zfa 1819	-11.58	-9.41
ag 440d 12	-10.79	-9.31
at 1zf6 1516	-7.82	-7.90
at 1zf6 56	-9.08	-6.47
ca 1zf6 45	-2.29	-0.28
ca 1zf8 23	-4.93	-2.33
cc 137d 67	-4.04	-2.15
cc 137d 78	-2.32	-3.07
cg 137d 23	-9.06	-9.37
cg 137d 89	-8.87	-10.46
ct 440d 1920	-5.02	-9.81
ct 440d 910	-6.56	-4.80
ga 1zfa 1718	-7.78	-8.27
gc 137d 56	-17.27	-14.89
gc 137d 910	-17.28	-14.24
gg 137d 34	-5.23	-4.19
gt 1zf8 78	-9.30	-8.13
gt 2pkv 23	-9.38	-7.69
ta 2pkv 34	-5.93	-3.64
ta 2pkv 56	-3.95	-2.89
tc 1zfa 34	-6.83	-5.86
tg 1zf6 67	-1.11	-1.45
tg 1zf8 1819	-6.18	-5.17
tt 1ikk 1314	-4.31	-4.24
tt 1ikk 1415	-4.59	-4.36
Average	-7.24	-6.46
BDNA		
aa 1d49 1617	-8.87	-5.29

aa_1d49_67	-8.54	-3.91
ac_1zf7_78	-3.74	-3.28
ag_196d_78	-11.54	-10.73
ag_1zfg_45	-9.68	-8.67
at_1d49_78	-7.38	-8.14
at_1d56_56	-7.49	-8.22
ca_1dc0_910	-8.11	-5.46
cc_1zf0_12	-2.90	-0.76
cc_1zf7_1112	-2.85	-0.95
cg_196d_56	-9.21	-10.71
cg_1d56_1920	-10.16	-12.90
ct_196d_34	-8.81	-5.56
ct_1zfg_1617	-5.05	-7.80
ga_196d_67	-15.05	-10.23
ga_196d_89	-13.54	-9.61
ga_1d49_23	-15.07	-9.09
gc_1fq2_2223	-19.13	-13.11
gc_1fq2_23	-17.97	-12.54
gg_1zf0_1920	-2.53	-5.07
gt_1zf0_1314	-7.94	-7.08
gt_1zf0_34	-7.81	-6.62
ta_1d49_56	-7.27	-7.05
ta_1d56_45	-6.52	-5.06
tc_196d_23	-5.80	-5.74
tc_196d_45	-6.62	-5.12
tt_1d49_1415	-5.42	-5.32
tt_1d49_45	-4.24	-6.36
Average	-8.54	-7.16
ADNA IDEAL Planar		
aa_adna_planar	-6.25	-5.46
ac_adna_planar	-4.80	-6.42
ag_adna_planar	-7.98	-6.71
at_adna_planar	-2.60	-6.26
ca_adna_planar	-4.73	-4.11
cc_adna_planar	-3.10	-2.14
cg_adna_planar	-6.38	-9.41
ct_adna_planar	-3.20	-4.32
ga_adna_planar	-9.32	-10.18

gc adna planar	-11.80	-13.91
gg adna planar	-3.86	-2.99
gt adna planar	-1.24	-6.99
ta adna planar	-5.78	-5.59
tc adna planar	-5.12	-6.84
tg adna planar	-3.97	-3.56
tt adna planar	1.89	-1.89
Average	-4.89	-6.05
BDNA IDEAL Planar		
aa bdna planar	-8.96	-5.39
ag bdna planar	-9.65	-8.05
ca bdna planar	-5.51	-4.07
cc bdna planar	-2.98	-1.67
cg bdna planar	-7.35	-10.55
ct bdna planar	-7.11	-6.33
ga bdna planar	-10.88	-10.18
gg bdna planar	-6.08	-4.63
ta bdna planar	-6.49	-5.79
tc bdna planar	-5.29	-5.49
tg bdna planar	-6.05	-6.03
tt bdna planar	-5.50	-4.15
Average	-6.82	-6.03

References

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2. McDonald, A.R., Denning, E.J. & MacKerell, A.D., Jr. Impact of geometry optimization on base-base stacking interaction energies in the canonical A- and B-forms of DNA. *J. Phys. Chem. A* **117**, 1560-1568 (2013).